GUEST TALK · ISA 387: DESIGNING BUSINESS SYSTEMS · PROF. SKIP BENAMATI

The FSB Research Atlas

Designing a Business System with Claude Code

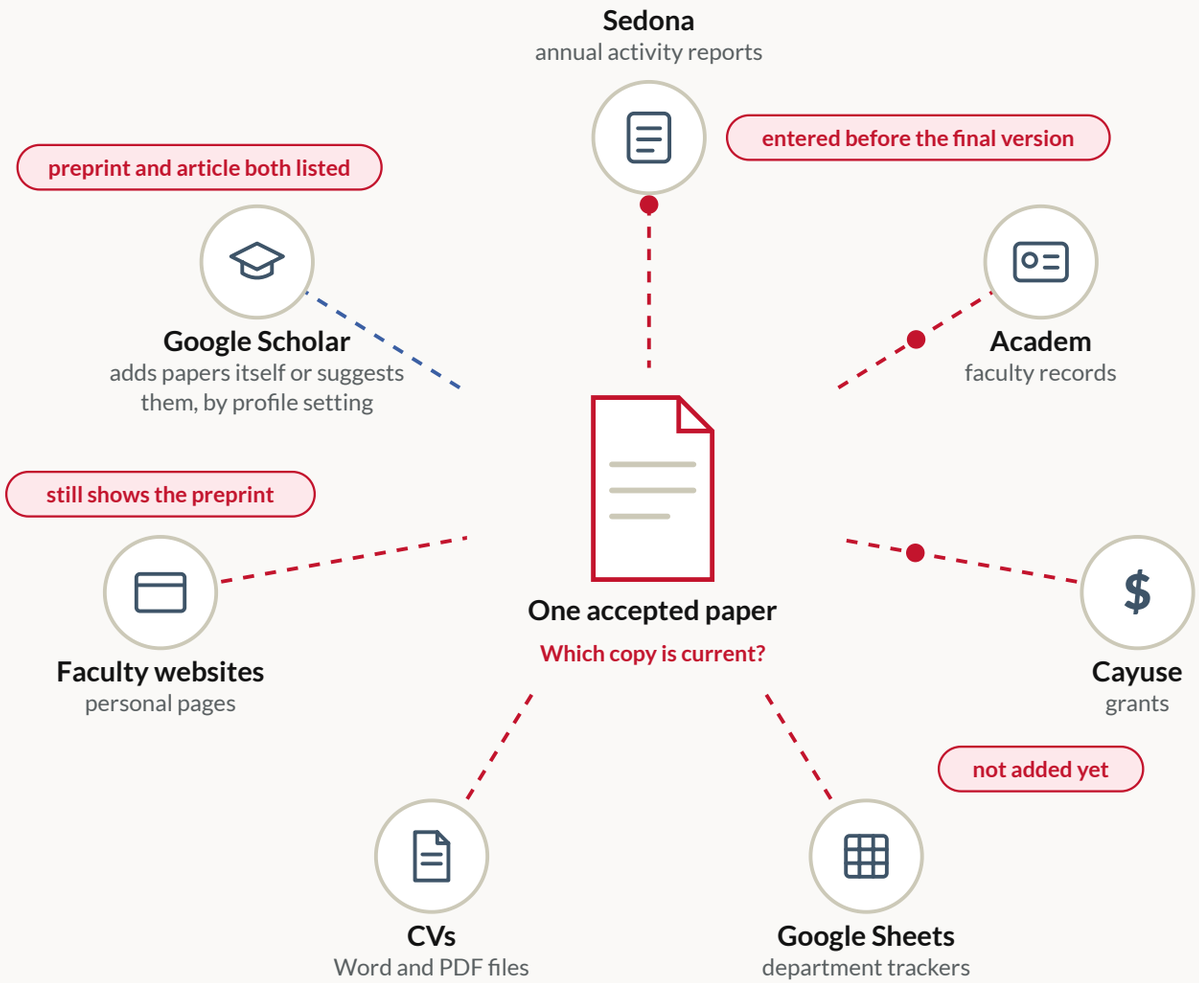
Fadel M. Megahed
Farmer School of Business, Miami University
fmegahed@miamioh.edu · October 21, 2026

Source code: github.com/fmegahed/fsb-research-atlas





The Dilemma: One Record in Many Systems



The same records live in many places.
Sedona, Academ, Cayuse, Google Sheets, CVs, faculty websites, and Google Scholar profiles each hold part of the picture.

Every new item is entered again, in each one.
An accepted paper, a grant, or an award is typed into most of these systems by hand. Google Scholar adds papers itself or suggests them for approval, depending on the profile settings.

Tracking becomes difficult.
Each copy is updated at a different time, so the copies disagree and nobody can say which one is current.

The Dilemma: Requests and Accreditation

Points of Pride Archived ×

Convert to PDF

Farmer, L. Allison
to FSB

Wed, Oct 14, 5:24 PM (6 days ago)

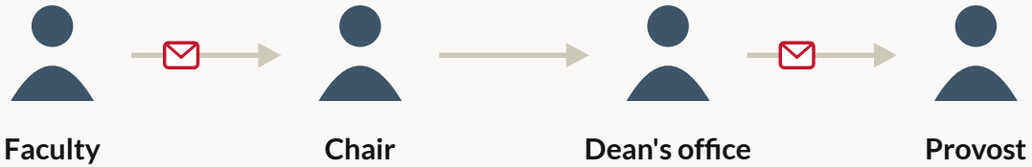
Dear Colleagues,

Each month, we report departmental accomplishments to the Dean's office. These go to the Provost. If you have any recent accomplishments not already reported, such as an accepted paper, a conference presentation, an award, a grant, etc., please let me know.

Thank you all!
Allison

A request sent to faculty on October 14, 2026 (signature removed).

EVERY MONTH



Each level collects the same accomplishments again and passes them up. Requests like this arrive on top of the regular reports.

EVERY ACCREDITATION REVIEW (AACSB)

Every intellectual contribution in the review window, by faculty member, type, and journal quality. For 2020 to 2025: 856 works.



122 of 145 tenure-line faculty published in the window.



Overview Video



0:00 Introduction **0:29** Public atlas **1:55** Numbers **2:09** Faculty app **3:10** Chairs **3:32** Dean's office **4:34** Governance **4:49** Closing

Recorded September 18, 2026. Some counts have changed since then after faculty review. Also on YouTube: youtu.be/73JrWLZGCwE

Key Numbers

THE SYSTEM

Public atlas, January 2023 to October 2026

614 research works by Farmer School faculty, 371 of them journal articles

167
active faculty

7
departments

4,695
citations to those journal articles

44
department themes in 8 school-wide areas

THE BUILD

August 21 to September 22, 2026

14 days with commits, working with Claude Code

245
commits, 237 of them co-authored by Claude

21,021
lines of code

38
automated test files

574
AI paper summaries, each editable by its authors

Sources: config/units/*/faculty.json, data/units/*/works.json, data/school/atlas.json, data/shared/summaries, and the git log of the repository at the October 7, 2026 refresh.

System Architecture

SOURCES

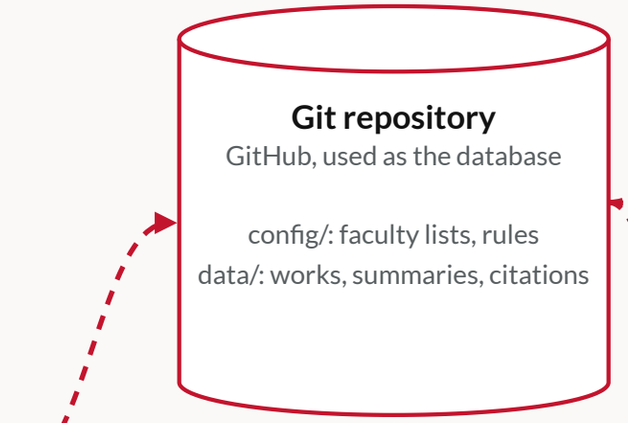
- **OpenAlex**
  works, authors, venues
- **Semantic Scholar**
  citations, abstracts
- **Crossref**
  DOIs, abstracts
- **Google Scholar via SerpApi**
  citation counts
- **NSF and NIH**
  federal grants
- **Open-access PDFs**
  full text for summaries
- **Wayback Machine**
  past faculty directories

PIPELINE

Monthly pipeline
TypeScript, src/refresh.ts

1. Roster: who is on the faculty
2. Fetch works from OpenAlex
3. Check uncertain author matches — AI
4. Add venues, abstracts, grants
5. Collect citations
6. Reconcile sources, remove duplicates
7. Download open-access full text
8. Summarize each new paper — AI
9. Assign research themes — AI
10. Verify, then build the site

STORAGE



GitHub Actions
monthly refresh, hourly fold, rebuild

OUTPUTS

Public website
static HTML on Vercel
fsbresearch.vercel.app

Signed-in app
Next.js, Google sign-in,
roles for faculty, chairs, deans

Reports
AACSB and annual reports,
Quarto, exported to Word

Monthly podcast
AI-written script,
ElevenLabs voices

Design choice: JSON files in git serve as the database. Every change is versioned, reviewable, and reversible, and the public site needs no database server.

Data Sources

SOURCE	WHAT IT PROVIDES	ACCESS
OpenAlex	Works, authors, institutions, venues	Free API key
Semantic Scholar	Citation counts, abstracts	Free API key
Crossref	DOIs, abstracts, publication metadata	Free API
Google Scholar	Citation counts from faculty profiles	SerpApi, $75 per month
NSF and NIH	Federal grants	Free APIs
Open-access full text	PDFs for the AI summaries	arXiv, Europe PMC, publishers
Wayback Machine	Faculty directory for past years	Free

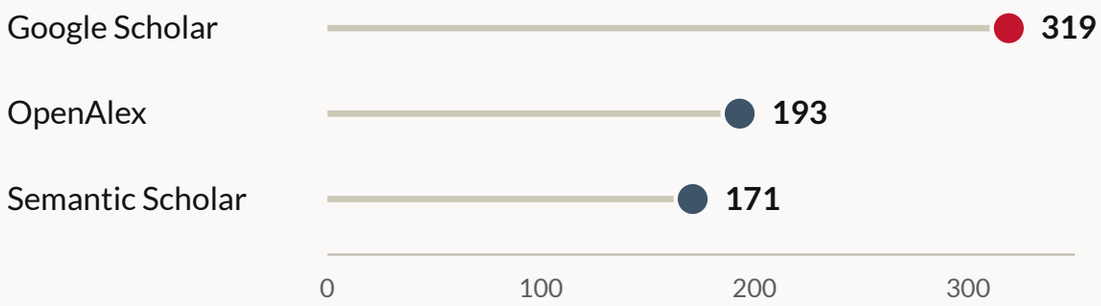
DATA PROBLEMS THE PIPELINE HANDLES

- OpenAlex merges different people who share a name.
- arXiv records list some authors as "Last, First".
- AIS eLibrary labels every item "Journal of the AIS".
- One paper can appear as a preprint, a dataset, and an article.

ONE PAPER, THREE CITATION COUNTS

How generative AI models such as ChatGPT can be (mis)used in SPC practice, education, and research? An exploratory study

Quality Engineering, 2023 · counts as of October 7, 2026



Rule: show Google Scholar when available, then Semantic Scholar, then OpenAlex, and label the source. The public page shows it:

Citations 319 (Google Scholar, as of 2026-10-07) · FWCI 38.09

Also: Semantic Scholar 171, OpenAlex 193

Where AI Is Used

Generated by AI

- Plain-language and technical summary of each paper (gpt-5.6-luna)
- Department themes and the eight school-wide areas
- A check on uncertain author matches
- AACSB category (basic, applied, teaching) and UN SDG tags
- Draft narrative for reports
- Monthly podcast script (gpt-5.6-sol) and voices (ElevenLabs)
- Theme artwork (gpt-image-1.5)

Computed by code

- Counting works and faculty
- Removing duplicate records
- Choosing the citation source
- Applying the reporting window and journal lists
- Building report tables
- Verifying output before anything is published

> Average cost of one AI paper summary: **$0.003**

Reviewed by faculty



Faculty edit any summary or tag and preview the public page before saving.

Automation and Governance

Monthly refresh

GitHub Actions, 7th of each month

1. **Scheduled run** — starts on its own
2. **Fetch** — new works and citations
3. **AI summaries** — new papers only
4. **Checks** — stop the run if counts look wrong
5. **Commit and deploy** — data to git, site to Vercel
6. **Digest issue** — what changed this month

Faculty edit

signed-in app

1. **Faculty member edits** — a summary, tags, or a wrong paper
2. **Decision record** — saved in Postgres: who, what, why
3. **Hourly fold job** — writes it to config/ as a commit
4. **Rebuild** — site, app data, and reports
5. **Live** — about 15 to 20 minutes later

git log · September 29, 2026 · author: atlas-app

```
app: 7 decisions (7 disclaim-work)

- removed: W4392713307 from [faculty] (finance) by [faculty]
- removed: W4385210481 from [faculty] (finance) by [faculty]
  ... 5 more

Decision-Id: 87285c1e-3e71-49d4-a477-03143b7de395
```

GOVERNANCE RULES

- Every number traces to a source record.
- Every edit is logged and can be reverted.
- The app writes only to Postgres; only the fold job writes to git.
- Miami Google sign-in; what you see depends on your role.
- No charts that rank departments against each other.

Development Process

Requirements
question-and-answer sessions with Claude before any code

Design
5 written specs, 1,308 lines

Plan
5 task-by-task plans, 2,505 lines

Build
Claude Code writes the code; I review each change

Test
38 test files, written before the code they check

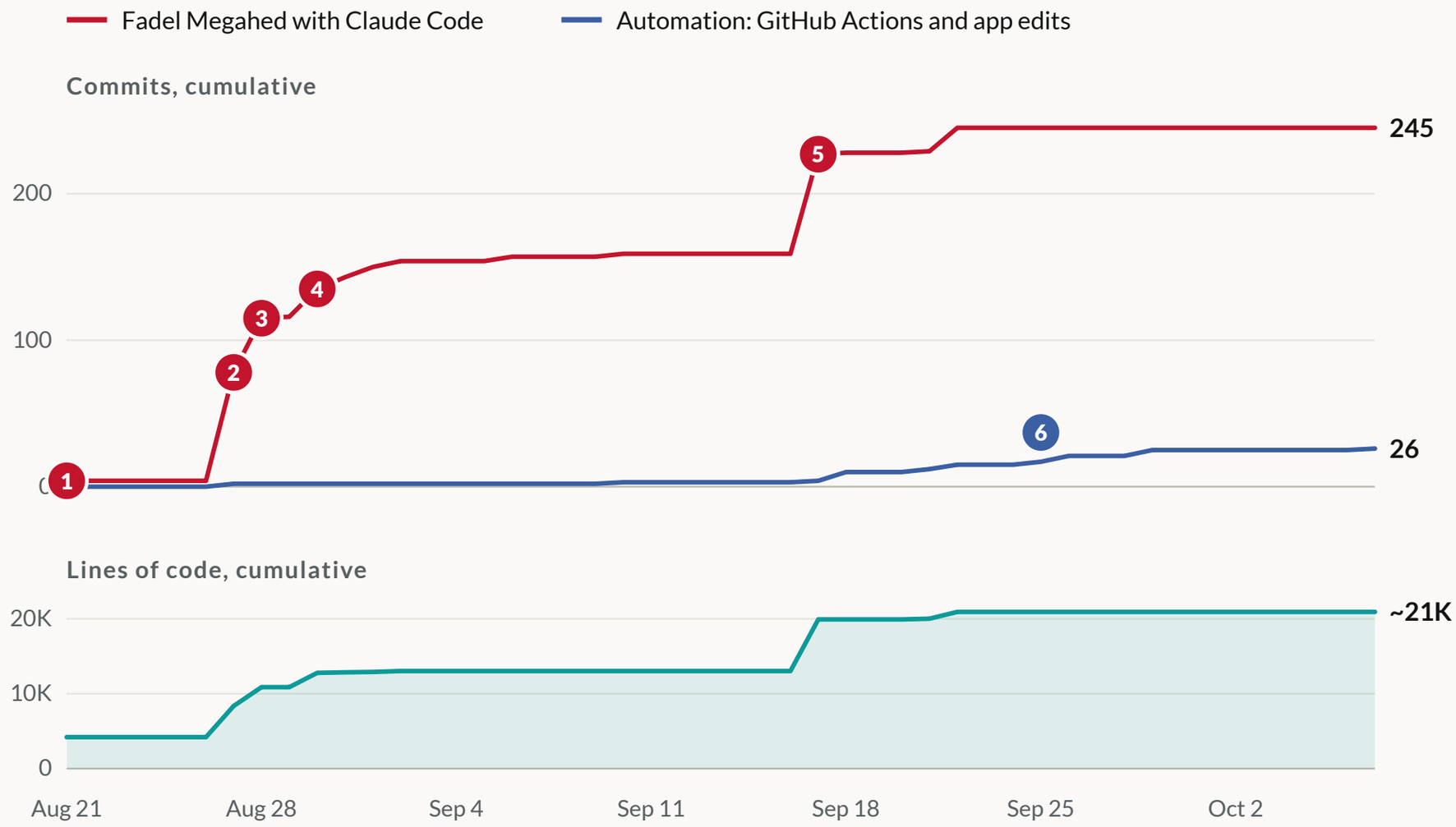
Maintain
CLAUDE.md project rules, 3 scheduled workflows

Plan · docs/superpowers/plans/2026-09-22-atlas-app.md

```
## Global Constraints
- Git stays source of truth: the app never writes repo files at runtime; only `app-fold.yml` commits.
- Domain restriction `miamioh.edu` enforced server-side on every session and every API route.
- Licensed AJG/JCR raw files never served; derived per-work columns only.
- No em dashes in user-facing copy.

### Task 3: Auth and roles
Steps: failing role tests (one per role + non-miamioh rejection) -> implement -> pass -> commit.
```

Development Timeline



1. **Aug 21** — ISA pipeline and website
2. **Aug 27** — Citations from three sources, monthly refresh, school website
3. **Aug 28** — All seven departments, podcast
4. **Aug 30** — AACSB and annual reporting
5. **Sep 17** — Signed-in app: spec, plan, build, and launch in one day
6. **Sep 25** — From here on, only automation commits

Costs

TO BUILD · AUGUST 21 TO SEPTEMBER 22

about $202

plus 14 days of my time

Claude Max 20x, one month **(shared across all my projects)**	**$200**
OpenAI API, all logged calls to date **(at least)**	**$2.27**

PER AI PAPER SUMMARY

$0.003

Every AI call records its token cost.

TO RUN · ABOUT $118 PER MONTH



$0: Neon Postgres (free tier), GitHub Actions (included minutes), and the OpenAlex, Semantic Scholar, Crossref, NSF, and NIH APIs.

Lessons Learned

1 **Write the design before the code**

Five specs and five plans set the scope before Claude wrote code. CLAUDE.md kept the project rules in place from one session to the next.

2 **Check AI and source data against the original**

OpenAlex merges different people who share a name. An AI check flags weak matches, and faculty remove wrong papers in the app.

3 **Treat early numbers as drafts**

The video reports 917 works for the 2020 to 2025 AACSB cycle. After a Google Scholar authorship review, the count is 856.

4 **Put checks into automation**

The monthly run stops when its checks fail, and every change is a git commit that can be reverted.

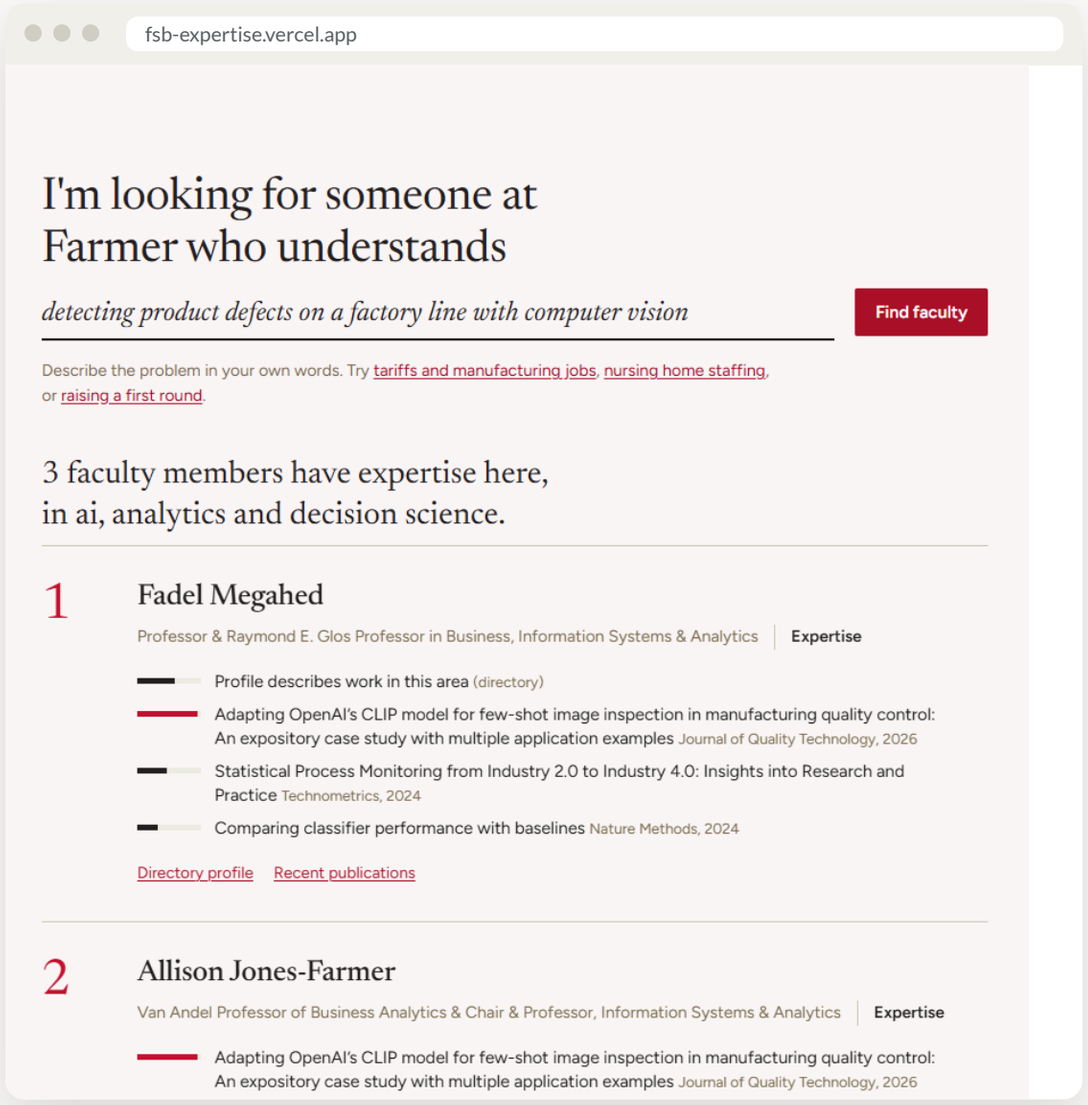
5 **Get feedback from users early**

The Research Committee reviewed the app in September. Their comments became GitHub issues #11 to #21, worked through on September 22.

6 **Track costs from the first day**

Every AI call records its token cost, so spending per paper and per month is known, not estimated.

Extension: Expertise Finder



A SECOND SYSTEM BUILT ON THE ATLAS DATA

Companies, media, and collaborators describe a problem in their own words and see the one to five Farmer School faculty with that expertise, each backed by specific papers or a public profile. If nobody matches, the page says so.

It reads the Atlas's records: 132 faculty, 693 faculty-paper links from journal articles since 2020, and public directory profiles.

HOW IT DECIDES

1 · Shortlist
one AI request reads the whole roster and picks finalists

2 · Read the evidence
one request per finalist rates each paper and the profile

3 · Code ranks
at most five faculty; only papers above a relevance threshold

> The AI model rates; it does not write text. Each answer is a typed judgment (a choice, a yes or no, or a score) from TypeSafe's Jev model.

1 day
to build: September 21, 9 commits

$0.002
per search (approximate)

1 to 2 s
per search

fsb-expertise.vercel.app

Building on This Project

github.com/fmegahed/fsb-research-atlas

Live site

Video

Expertise finder

HOW TO START

1. Clone the repository; read `README.md` and `CLAUDE.md`.
2. Read one spec in `docs/superpowers/specs/` to see how a feature was designed.
3. Run `npm install` and `npm test`.
4. Open Claude Code in the folder and ask it to explain the architecture.

IDEAS FOR EXTENSIONS

- Adapt it to another school or university
- Track grants and funding over time
- An accessibility review of the public site
- Add student research and capstone projects
- A chat assistant over the atlas data
- Better author disambiguation

Questions?